

Intrinsic origin of room temperature ferromagnetic ordering in $(\text{In}_{0.97-x}\text{Co}_x\text{Mg}_{0.03})_2\text{O}_3$ thin films

Jiashuo Yan^a, Cunquan Li^a, Huijuan Sun^{b,*,**}, Yukai An^{a,*}

^a Key Laboratory of Display Materials and Photoelectric Devices, Ministry of Education; Tianjin Key Laboratory for Photoelectric Materials and Devices; School of Material Science and Engineering, Tianjin University of Technology, Tianjin 300384, China

^b Institute of Mathematics and Physics, Beijing Union University, Beijing 100101, China

ARTICLE INFO

Handling Editor: Prof. L.G. Hultman

Keywords:

In_2O_3
Local structure
Magnetic properties
Transport properties

ABSTRACT

The $(\text{In}_{0.97-x}\text{Co}_x\text{Mg}_{0.03})_2\text{O}_3$ films with different Co concentration are prepared using RF magnetron sputtering. The XRD and XPS measurements show that the films are highly oriented and both Co and Mg elements present valence state of +2. The EXAFS spectra show that only substitutional Co ions exist in the film with $x = 0.02$. However, the films with higher Co concentration ($x > 0.02$) form separate phases of substitutional Co and metallic Co clusters. All the films exhibit n-type conductivity and the carrier concentration firstly decreases and then increases with Co doping. The transport mechanisms are dominated using Mott variable range hopping at low temperature and Hard bandgap hopping at high temperature. The Co dopant can effectively alter the MR characters, which undergoes a clear transition from negative MR to positive MR. All the films exhibit room temperature ferromagnetic hysteresis loop. Clear increase of ferromagnetic hysteresis loop and saturation magnetic moment are observed due to the incorporation of Co ions. The ferromagnetism is ascribed to oxygen vacancies-induced bound magnetic polarizations. The relative percentage of substitutional Co ions decrease with Co doping, which leads to the decreasing of number of bound magnetic polarons induced by oxygen vacancies, and then weaken the Ms.

1. Introduction

For the traditional oxide semiconductors of ZnO [1–4], SnO_2 [5–7] and In_2O_3 [8–12], d orbitals of their metal cations are in a fully filled state and cannot provide a magnetic moment. If the ferromagnetic semiconductors doped by 3d or 4f transition metals (TM) can be successfully prepared, it is very promising to make them into multifunctional devices integrating magnetism-electricity-light-gas. Since the theoretical prediction of Mn doped ZnO with room temperature (RT) ferromagnetic ordering [13], the wideband gap oxide semiconductors gradually become a hot topic [14]. In view of the importance of magnetic element doping in transparent conductive oxide semiconductors, numerous researchers have conducted research on related topics. In_2O_3 with wide-band gap of 3.75 eV exhibits potential applications, such as, liquid crystal displays, flat panel displays and chemical sensors, due to their many advantages of both visible light transparency, high conductivity, and excellent gas-sensitive properties [15].

The relevant investigations on physical properties of In_2O_3 -based

diluted magnetic semiconductors (DMS) by TM doping or co-doping have been carried out. Duan et al. studied the spin-dependent magnetic behaviors of Co-doped In_2O_3 thin films, and considered that the ferromagnetic ordering is not from the Co clusters, but more due to the intrinsic effect of substitutional Co^{2+} [16]. Similarly, Gloria et al. found the precipitation of Co clusters in the Co-doped In_2O_3 films, and concluded that the source of ferromagnetism at room temperature is due to the presence of second magnetic phases [17]. Jiang et al. deposited the Fe/Cu co-doped In_2O_3 films with RT ferromagnetism, and found that the non-magnetic Cu doping can lead to the appearing of Fe metallic clusters [18]. In view of these, TM doping or co-doping in oxide DMSs is undoubtedly an important means to obtain RT intrinsic ferromagnetism and uniform distribution of doped magnetic atoms in thin film samples. The observed ferromagnetism in doped In_2O_3 systems possibly originates from the magnetic impurity phases, which result an extrinsic magnetic mechanism. On the other hand, some intrinsic ferromagnetic mechanisms have suggested, including super-exchange magnetic interaction between magnetic ions, carrier-mediated magnetic interaction

* Corresponding author.

** Corresponding author.

E-mail addresses: ldthuijuan@bnu.edu.cn (H. Sun), ykan@tjut.edu.cn (Y. An).

<https://doi.org/10.1016/j.vacuum.2024.113102>

Received 18 September 2023; Received in revised form 1 March 2024; Accepted 2 March 2024

Available online 2 March 2024

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and bound magnetic polaron (BMP) mechanism based on structural defects (such as oxygen vacancy), etc. Therefore, to clarify the magnetic origin or possible magnetic interactions in the doped or co-doped In_2O_3 systems, it is important to understand how the magnetic dopants adjust and alter the spin-dependent magnetic and transport properties of In_2O_3 based DMs films, in accompany with the change of structural defects.

In this paper, the Mg/Co co-doped In_2O_3 films with different Co concentrations are prepared by magnetron sputtering technology. The X-ray absorption spectra at Co K-edge are measured to determine the local structures (occupation site, Co metal cluster and Co oxide) of doped Co atoms. The corresponding physical properties, such as carrier concentration, ρ -T curves, hysteresis loop and magnetoresistance (MR) of Mg/Co co-doped In_2O_3 films are investigated in detail, which helps to clarify the correlation among the oxygen vacancies, Co dopant and magnetic coupling.

2. Experimental details

The $(\text{In}_{0.97-x}\text{Co}_x\text{Mg}_{0.03})_2\text{O}_3$ films are prepared using the FJL560 ultra-high vacuum opposite-target magnetron sputtering system. The base pressure of system is about 8.5×10^{-5} Pa. During deposition, the substrate temperature is kept at 400°C with RF power of 90W, and the Ar and O_2 flow rate are controlled at 10 sccm and 0.6 sccm, respectively. Pure In_2O_3 ceramic target is selected as the sputtering material. Regularly shaped Co sheets (99.99% purity) and MgO sheets (99.99% purity) are symmetrically arranged on the surface of target. By altering the number of Co sheets, the doped In_2O_3 films with various Co concentration are deposited on the SiO_2/Si (100) substrates. The thickness of films is about 550 nm, which is determined using KLA Tencor Alpha Step D-100 surface profiler. The nonstoichiometric formula for the Co/Mg codoped In_2O_3 films are determined to be $(\text{In}_{0.97}\text{Mg}_{0.03})_2\text{O}_{2.74}$, $(\text{In}_{0.95}\text{Co}_{0.02}\text{Mg}_{0.03})_2\text{O}_{2.7}$, $(\text{In}_{0.93}\text{Co}_{0.04}\text{Mg}_{0.03})_2\text{O}_{2.69}$, $(\text{In}_{0.9}\text{Co}_{0.07}\text{Mg}_{0.03})_2\text{O}_{2.68}$, $(\text{In}_{0.885}\text{Co}_{0.085}\text{Mg}_{0.03})_2\text{O}_{2.66}$ and $(\text{In}_{0.87}\text{Co}_{0.1}\text{Mg}_{0.03})_2\text{O}_{2.67}$ by energy dispersive X-Ray spectroscopy (EDS). The crystal structure is detected by the Rigaku Dmax2500PC X-ray diffraction ($\lambda = 0.15406$ nm). X-ray photoelectron spectroscopy (XPS) is used to analyze the chemical valence of films by PHI-1600 ultrahigh vacuum (1×10^{-9} bar) unit with a focused monochromatic Mg K α ($h\nu = 1253.6$ eV). The surface of films is firstly sputter-etched for 30 s with monatomic Ar^+ at an energy of 1 keV. Moreover, a charge neutralizer is used throughout the measurements and the spectra of the elements In, O, Mg and Co are measured sequentially. The XPS spectrum is calibrated with reference to C1s set at $289.58\text{-}\Phi_{\text{SA}}$, and the work function Φ_{SA} of In_2O_3 is about 4.3 eV [19]. Therefore, the binding energy of the C1s peak of adventitious carbon in In_2O_3 is calculated to be 285.28 eV, which is used as the reference peak. The Co K-edge X-ray absorption spectra (XAS) are tested

using total fluorescence mode at the diffraction station (4B9A) of Beijing Synchrotron Radiation. The Co L-edge XAS spectra are tested using total electron-yield (TEY) mode at the soft x-ray station (4B7B). During the measurements, the incident X-ray is fixed to 45° relative to the surface of the films. The incident beam is collimated by slits and monochromatized with Si (111) crystal, and the photon energy is calibrated by measuring a metallic Co foil. The energy resolution $\Delta E/E$ is estimated to be 2×10^{-4} . The hysteresis loop of films is carried out on a Qunyum Design PPMS-9 physical property tester. The HMS-505 Hall effect tester is used to measure the carrier concentration, conductive type and resistivity of the films by van der Pauw method. Using the RTM-505 low temperature transport test system, the ρ -T and M-R curves is measured by the four-probe method.

3. Results and discussion

XRD curves for the $(\text{In}_{0.97-x}\text{Co}_x\text{Mg}_{0.03})_2\text{O}_3$ films with various Co concentration are displayed in Fig. 1(a). The diffraction peaks are sharp and well match with the standard pattern of In_2O_3 (space group of $\text{Ia}\bar{3}$) [20]. No diffraction peaks of impurity phases, such as metallic Co clusters and oxides are found. The inset is the magnified images of (222) peaks. Clearly, the diffraction peaks move towards a higher angle when the Co doping concentration $x \leq 0.04$, implying a gradual decrease of lattice spacing. According to the Bragg equation $2d \sin \theta = n\lambda$, the decrease of lattice constant is due to that the smaller radius of Co ions compared to the In^{3+} ions ($0.94 \text{ \AA}$). When the Co doping concentration $x > 0.04$, the position of (222) peak move towards a lower angle. Possibly, the forming of Co metallic clusters makes the percentage of substitutal Co ions decrease, resulting that the lattice constant becomes larger compared to the films with low Co concentration. The XRD pattern is not sensitive to the existence of metallic cluster at nanoscale. Therefore, the presence of the impurity phases in the film with high Co doping concentrations cannot be accurately determined, and further confirmation is still needed in combination with other characterization methods. Fig. 1(b) shows the lattice parameters a with respect to Co concentration x for the $(\text{In}_{0.97-x}\text{Co}_x\text{Mg}_{0.03})_2\text{O}_3$ films. The lattice parameter a firstly decreases and then increases with Co doping. The above analysis strongly suggest that when the Co concentration $x \leq 0.04$, the doped Co atoms mainly enter the In_2O_3 lattice in the form of substitution. The increase of substitutal Co ions lead to the decrease of lattice parameters. When the Co concentration $x > 0.04$, Co metallic clusters gradually precipitate in the films, resulting in the decrease of substitutal Co ions and the increase in lattice parameters.

Fig. 2(a) shows the HRTEM image of $(\text{In}_{0.85}\text{Co}_{0.1}\text{Mg}_{0.03})_2\text{O}_3$ film. The corresponding magnified images of regions (1)–(3) in Fig. 2(a) are displayed in Fig. 2(b)–(d), respectively. The streak spacing of region (1) in

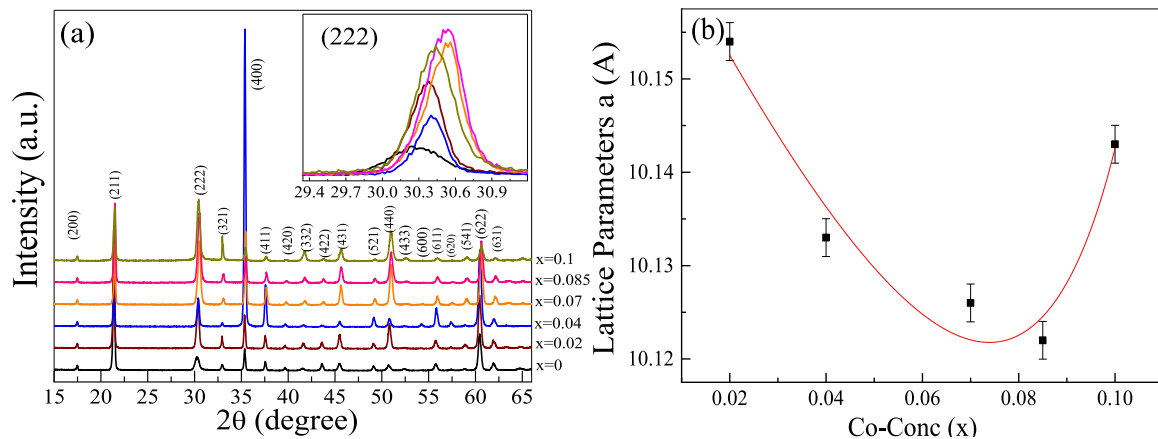


Fig. 1. (a) The XRD curves for the $(\text{In}_{0.97-x}\text{Co}_x\text{Mg}_{0.03})_2\text{O}_3$ films. The inset is the magnified images of (222) peaks. (b) The lattice parameters a with respect to Co concentration x .

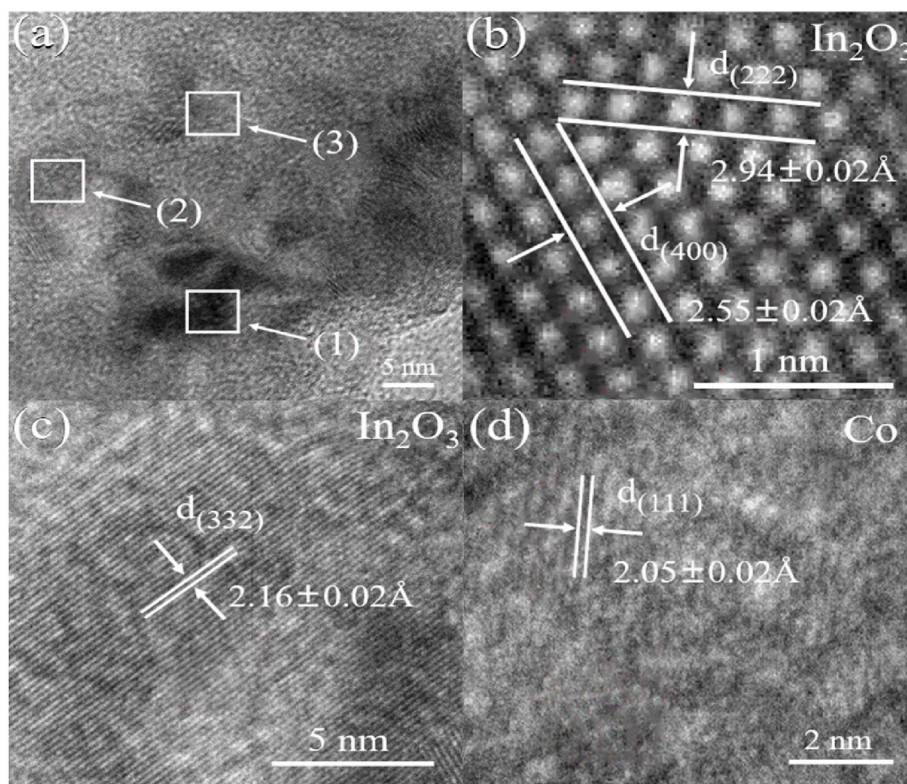


Fig. 2. (a) The HRTEM image of $(\text{In}_{0.85}\text{Co}_{0.1}\text{Mg}_{0.03})_2\text{O}_3$ film. (b–d) The enlarged HRTEM views of region (1), region (2) and region (3) in Fig. 2(a).

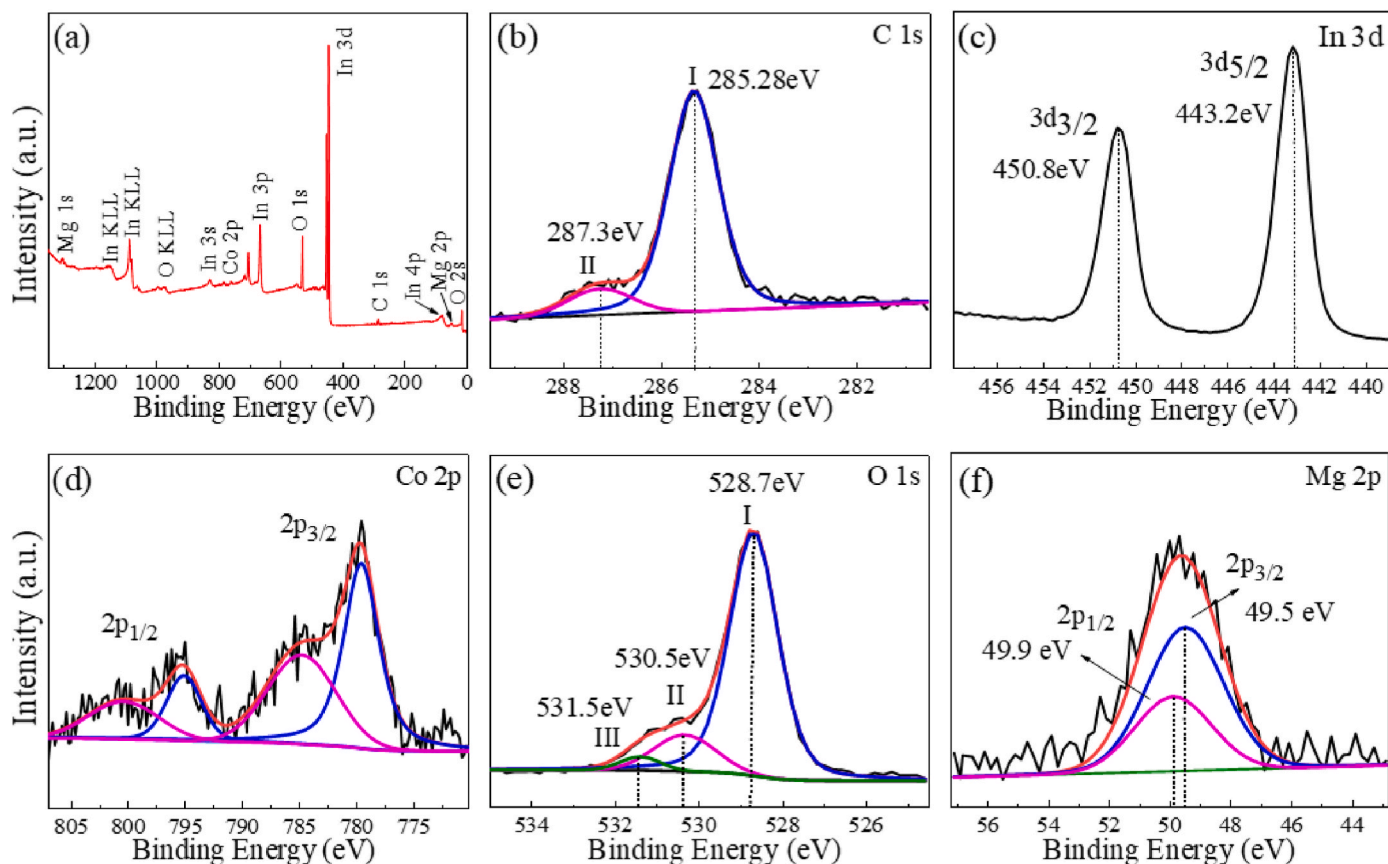


Fig. 3. (a) XPS survey spectrum as well as (b) C 1s, (c) In 3d, (d) Co 2p, (e) O 1s and (f) Mg 2p XPS spectra for the $(\text{In}_{0.93}\text{Co}_{0.04}\text{Mg}_{0.03})_2\text{O}_3$ film.

Fig. 2(a) is clearly discernible, showing an average lattice spacing of 2.94 Å and 2.55 Å with a crystal plane angle of 56.86°. The corresponding interplanar indices are (222) and (400), with the theoretical d values of 2.92 Å and 2.52 Å, and theoretical angle of 54.79°, which are consistent with the experimental values within the error range. However, the streak spacing of region (3) enlarged in Fig. 2(d) is not as clear as the regions (1) and (2). Its lattice spacing is determined to be 2.05 Å, which well matches the (111) crystal plane of metallic Co with hexagonal structure.

Fig. 3(a) shows the XPS survey spectrum for the prepared $(\text{In}_{0.93}\text{Co}_{0.04}\text{Mg}_{0.03})_2\text{O}_3$ film. Clearly, there exist C, O, In, Co and Mg elements in the surface of film. To understand the elemental valence of film, the C 1s, In 3d, Co 2p, O 1s and Mg 2p XPS spectra of film are carried out, as shown in Fig. 3(b)–(e). The C 1s XPS peak in Fig. 3(b) can be deconvoluted into two characteristic peaks, I and II, located at 285.28 eV and 287.3 eV, respectively. The peak I originates from the contribution of adventitious carbon and the peak II is usually associated with carbonyl C atoms [21]. The spin-orbit coupling effect causes the splitting of In 3d and Co 2p core levels and exhibits stronger contributions of In 3d_{5/2} and Co 2p_{3/2}, as shown in Fig. 3(c)–(d). The core levels of In 3d_{5/2} and In 3d_{3/2} are located at 443.2 eV and 450.8 eV, respectively, proving the existence of In³⁺ state [22,23]. The Co 2p_{3/2} and Co 2p_{1/2} peaks in Fig. 2(d) are located at 779.6 eV and 795.1 eV with energy difference of 15.5 eV, respectively, suggesting that the Co element in the film mainly exists as Co²⁺ ions [24]. Strong satellite characters at ~5.5 eV on the higher binding energy side of main peak can be also observed, implying the typical character of Co ions with high spin divalent state

[24,25]. The O 1s peak in Fig. 3(e) is asymmetric and is combined by peak I (528.7 eV), peak II (530.5 eV) and peak III (531.5 eV) after Gaussian fitting. The peak I is the binding energy of O²⁻ in In₂O₃, the peak II corresponds to the binding energy of O ion around the oxygen vacancies (V_O) and the peak III is associated with carbonyl O atoms [26,27]. The core levels of Mg 2p_{3/2} and 2p_{1/2} peaks in Fig. 3(f) are located at 49.5 eV and 49.9 eV with a small energy difference of 0.4 eV, corresponding to the Mg²⁺ state [28].

Fig. 4(a) displays the XANES spectra of $(\text{In}_{0.97-x}\text{Co}_x\text{Mg}_{0.03})_2\text{O}_3$ ($x = 0.02, 0.04, 0.07, 0.085, 0.1$) films and standard samples of Co, CoO, Co₃O₄ and Co₂O₃ measured at Co K-edge. Clearly, for the $(\text{In}_{0.97-x}\text{Co}_x\text{Mg}_{0.03})_2\text{O}_3$ film with $x = 0.02$, its absorption edge is very close to that of CoO, and a strong white line peak located in the higher energy side of the absorption is also observed, suggesting that the Co ions exists as +2 valence state. As the Co concentration x increases, the absorption edge gradually shifts to metallic Co, indicating that more Co atoms in the films are in the 0 valence state. Fig. 4(b) plots the dependence of threshold energy (E_0) on the Co oxidation state for the standard samples of Co, CoO, Co₃O₄ and Co₂O₃, which displays a linear increase between the E_0 and Co oxidation states. The E_0 for the $(\text{In}_{0.97-x}\text{Co}_x\text{Mg}_{0.03})_2\text{O}_3$ films with $x = 0.02, 0.04, 0.07, 0.085$ and 0.1 are 7714.78 eV, 7714.23 eV, 7713.78 eV, 7713.64 eV and 7713.37 eV, respectively, and for standard samples of Co and CoO are 7708.7 eV and 7715.33 eV, respectively. These again prove a mixed-valent state (Co⁰/Co²⁺) in the films with high Co concentration x . Fig. 4(c) is the EXAFS $k^3\chi(k)$ curves of $(\text{In}_{0.97-x}\text{Co}_x\text{Mg}_{0.03})_2\text{O}_3$ ($x = 0.02, 0.04, 0.07, 0.085, 0.1$) films and standard samples of Co, CoO, Co₃O₄ and Co₂O₃ measured at Co K-edge.

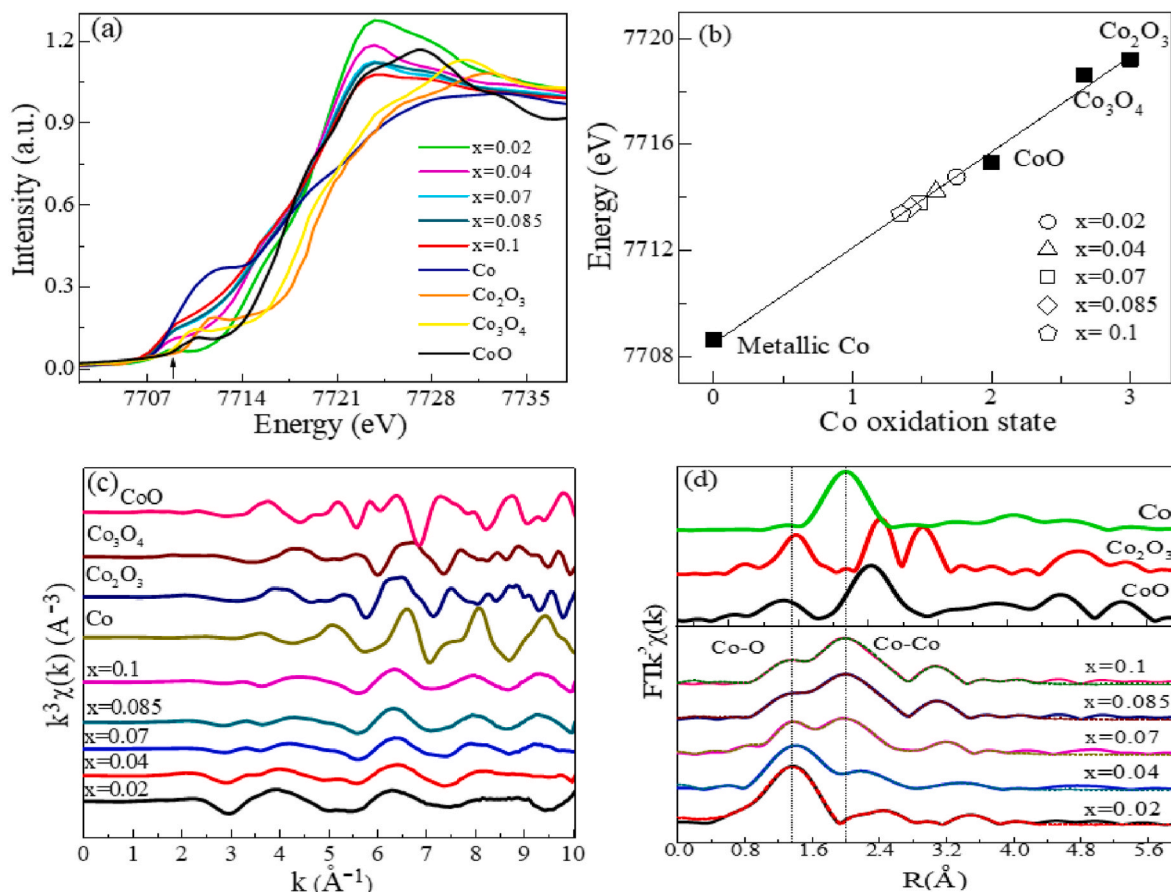


Fig. 4. (a) The XANES spectra for the $(\text{In}_{0.97-x}\text{Co}_x\text{Mg}_{0.03})_2\text{O}_3$ ($x = 0.02, 0.04, 0.07, 0.085, 0.1$) films and standard samples of Co, CoO, Co₃O₄ and Co₂O₃ measured at Co K-edge. (b) The dependence of threshold energy (E_0) on the Co oxidation state for the standard samples of Co, CoO, Co₃O₄ and Co₂O₃. The established dependence is used to infer the effective Co oxidation state in the $(\text{In}_{0.97-x}\text{Co}_x\text{Mg}_{0.03})_2\text{O}_3$ films (open symbols). (c) EXAFS $k^3\chi(k)$ curves and (d) their Fourier transforms for the $(\text{In}_{0.97-x}\text{Co}_x\text{Mg}_{0.03})_2\text{O}_3$ ($x = 0.02, 0.04, 0.07, 0.085, 0.1$) films and standard samples of Co, CoO, Co₃O₄ and Co₂O₃ measured at Co K-edge. The fitting curves of Fourier transforms are displayed by the dash lines.

One can see that the $k^3\chi(k)$ curve with $x = 0.02$ shows obvious different oscillation behavior compared with those of the standard samples, implying that the Co clusters and the second phase of impurities such as Co oxides can be excluded. However, for the films with $x > 0.02$, the characteristic peaks around $8 \text{ \AA}^{-1}$ and $9.37 \text{ \AA}^{-1}$ of $k^3\chi(k)$ oscillations are obvious. As the Co concentration increases, the intensity of peaks enhances, while the position of peaks is basically unchanged. This also indirectly proves that when the Co concentration increases to a certain threshold, a small amount of Co clusters will precipitate in the film and the content gradually increases. Fig. 4(d) shows the corresponding EXAFS Fourier transform (R-space) of $(\text{In}_{0.97-x}\text{Co}_x\text{Mg}_{0.03})_2\text{O}_3$ films and standard samples of Co, CoO and Co_2O_3 , where the dotted line is the fitting curve. It is noted that these peaks can be shifted by about $0.7 \text{ \AA}$ on the R axis from their true bond lengths due to the phase shift of the back scattered photoelectrons. Clearly, when $x = 0.02$, there exist three peaks at 1.36 , 2.35 and $3.47 \text{ \AA}$, corresponding to the Co–O, Co–In and Co–In neighbor coordination of Co dopant. When $x > 0.02$, a clear additional peak at $2.05 \text{ \AA}$ appears, which is very similar to that of metallic Co cluster at $2.12 \text{ \AA}$. With the increase of Co concentration x , the peak intensity for the Co–In coordination at $1.36 \text{ \AA}$ obviously become weak and the additional peak intensity ($2.05 \text{ \AA}$) becomes stronger, which can be attributed that the increase of Co metallic clusters reduces the content of substitutional Co. The Athena and Artemis interfaces to the IFEFFIT program package is used to fit the EXAFS Fourier transform curves (R-space and not the raw data) of films with window of $0\text{--}4 \text{ \AA}$, which includes the first, second and third coordination shells of Co dopant [29–33]. Generally, an ideal unit cell In_2O_3 includes 32 In atoms and 48 O atoms. Due to the presence of vacancies, two O^{2-} ions are removed from a cube with eight coordination points, resulting that the In ions have the same coordination number of 6 in the first (In–O), second (In–In) and third (In–In) coordination shells. Therefore, there exist two kinds of In atoms, namely In_1 atom (b-site) and In_2 atom (d-site), corresponding that two O^{2-} ions are missing at the top angle of the body diagonal and face diagonal, respectively. In the fitting process, the amplitude reduction factor S_0^2 of 0.90 determined from the Co foil is used for the Co K edges. Co replacement of In_1 site is firstly used as the model. It can be found that the fitted curves obtained from the model are in good agreement with the R-space curves for the film with $x = 0.02$. However, considering that Co metal clusters precipitated in films with $x > 0.02$, a combination model of Co substitution In_1 site and Co metal clusters with hexagonal closest packed (hcp) structure is selected for fitting. The corresponding fitted parameters are displayed in Table 1. As

Table 1

Fitted parameters of Fourier transforms of Co K-edge EXAFS $k^3\chi(k)$ functions. N , R and σ^2 are the coordination number, bond length and the Debye-waller factor, respectively. The statistical error for N , R , and σ^2 are 1%, $0.01 \text{ \AA}$, and 5%, respectively.

Sample	Neighboring atom	N	R($\text{\AA}$)	$\sigma^2(\text{\AA}^2)$
$x = 0.02$	O	5.72	2.165	0.0017
	In	5.72	3.232	0.0173
	In	5.72	3.654	0.0117
$x = 0.04$	O	5.64	2.151	0.0052
	In	5.64	3.228	0.0121
	In	5.64	3.625	0.0154
Phase: Co metal $x = 0.07$	Co	10.52	2.361	0.0194
	O	5.42	2.142	0.0072
	In	5.42	3.208	0.0156
Phase: Co metal $x = 0.085$	Co	5.42	3.605	0.0094
	O	10.83	2.373	0.0178
	In	5.13	2.134	0.0083
Phase: Co metal $x = 0.1$	In	5.13	3.258	0.0041
	In	5.13	3.567	0.0093
	Co	11.25	2.389	0.0103
Phase: Co metal	O	4.95	2.138	0.0151
	In	4.95	3.268	0.0200
	In	4.95	3.552	0.0048
Phase: Co metal	Co	11.64	2.397	0.0112

can be seen from Table 1, for the films with $x < 0.07$, the bond lengths of Co–O coordination are slightly smaller compared with the bond length ($2.19 \text{ \AA}$) of In–O coordination in In_2O_3 crystal. While as the Co concentration further increases, the bond length and effective coordination number N of Co–O coordination gradually decrease, which possibly correlates with the increase of substitutional Co ions. When the Co concentration $x \geq 0.04$, the effective coordination number N of Co–O coordination decreases sharply due to the precipitation of the Co cluster. On the contrary, the disorder factor σ^2 (Debye-waller factor [34–36]) of the first nearest neighbor (Co–O) shows a clear increase as the Co concentration increases. It can consider that the lattice distortion caused by Co doping distorts the Co–O coordination and leads to the increase of disorder. While the second nearest neighbor of Co– In_1 and the third nearest neighbor of Co– In_2 are less affected by this distortion, so the change of σ^2 is not very small. However, in fact, the σ^2 of the second and third nearest neighbors is much larger than that of the first nearest neighbor, which is probably because the Mg doping plays a certain superposition effect on the distortion of lattice. The fitting results also show that when the doping concentration $x \geq 0.04$, the Co–Co bond length in the film is still smaller than that ($2.5 \text{ \AA}$) of standard Co metal. It may be due to that when the Co doping concentration reaches a threshold value, some Co atoms precipitate out and form Co rich regions, which are distributed in a disordered manner in the films with smaller Co–Co spacing. In addition, with increasing the Co concentration x , the bond length of Co–Co and its coordination number gradually increases from $2.361 \text{ \AA}$ to $2.397 \text{ \AA}$ and from 10.52 to 11.64 , respectively, which are closer to those of standard Co metal. On the contrary, the disorder factor σ^2 shows a decreasing trend. These imply that the Co rich regions are distributed in a more ordered manner for the films with high x .

Fig. 5 shows the Co L-edge XAS of $(\text{In}_{0.97-x}\text{Co}_x\text{Mg}_{0.03})_2\text{O}_3$ films and standard samples of Co, CoO and Co_2O_3 . All experimental spectra exhibit sharp L_3 peaks (transition of Co $2p_{3/2}\text{--}3d$) and flat L_2 peaks (transition of Co $2p_{1/2}\text{--}3d$). The L-edge XAS is strongly related to the chemical valence of Co dopant. With increasing Co concentration, both the position of L_2 and L_3 peaks gradually shifts to the lower energy, which well matched with the energy of Co metal. Therefore, it can consider that for the film with $x = 0.02$, the Co atoms are mainly present in the substitution form. When $x > 0.02$, a small amount of Co clusters will be precipitated in the films, corresponding with the EXAFS results. Fig. 6(a)–(e) displays the Co $2p_{3/2}$ core level (L_3 peak) deconvoluted XAS

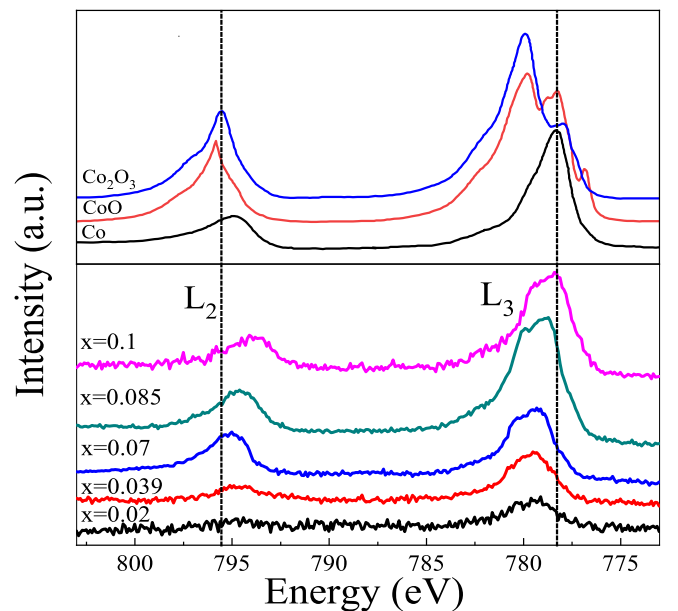


Fig. 5. The Co L-edge XAS of $(\text{In}_{0.97-x}\text{Co}_x\text{Mg}_{0.03})_2\text{O}_3$ films and standard samples of Co, CoO and Co_2O_3 .

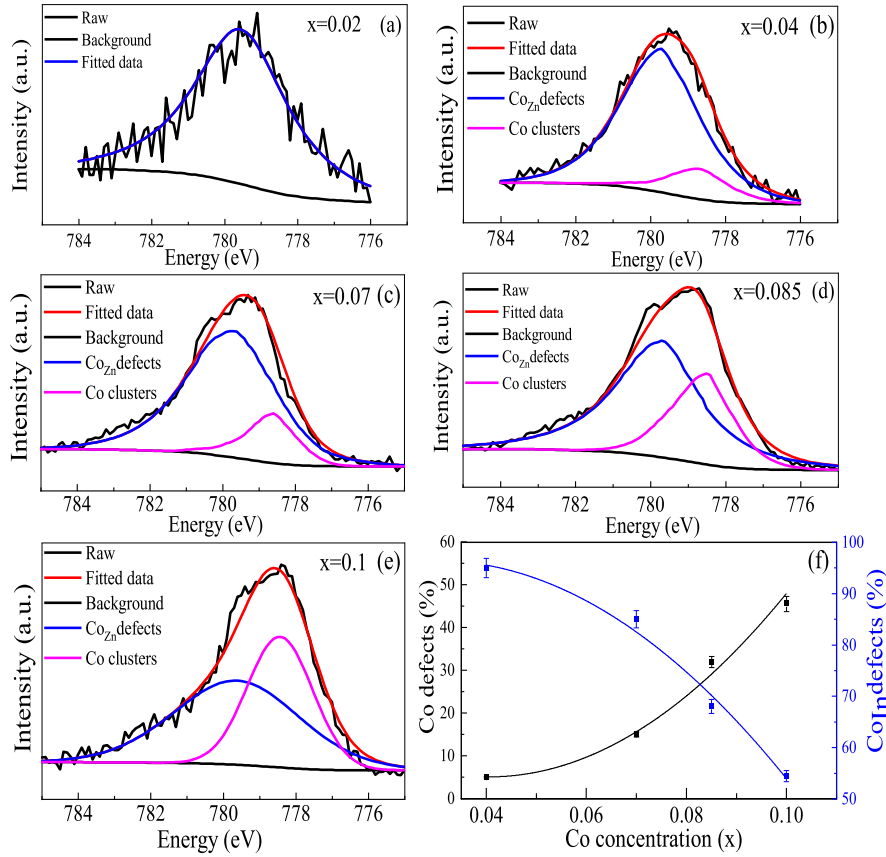


Fig. 6. Co $2p_{3/2}$ core level deconvoluted XAS with (a) $x = 0.02$, (b) $x = 0.04$, (c) $x = 0.07$, (d) $x = 0.085$ and (e) $x = 0.1$. (f) The area of deconvoluted peaks under various Co concentration.

under various Co concentration. Clearly, the Co $2p_{3/2}$ core level XAS spectra are obvious asymmetric except for the film with $x = 0.02$. The L_3 peak in Fig. 6(b)–(e) can be decomposed into two peaks of 779.9 and 778.6 eV, corresponding to the Co_{In} defect and metallic Co cluster, respectively. The percentage of metallic Co clusters gradually increases with Co doping, while the Co_{In} defects show the opposite trend, as shown in Fig. 6(f). After calculating the area ratio of two peaks, the percentage of metallic Co clusters with $x = 0.04, 0.07, 0.085$ and 0.1 is estimated to be 5 at%, 15 at%, 32 at% and 46 at%, respectively.

Fig. 7(a) shows the hysteresis loops of $(\text{In}_{0.97-x}\text{Co}_x\text{Mg}_{0.03})_2\text{O}_3$ films at 300 K. All the films exhibit a clear hysteresis loop, indicating the characters of room temperature ferromagnetic ordering. The magnified loops at low magnetic field also prove the existence of coercive field about 50–80 Oe, as shown in the top inset of Fig. 7(a). The bottom inset of Fig. 7(a) displays the M_s as a function of Co concentration x . Obviously, the M_s shows a nonmonotonic variation as the Co concentration x increases. According to the previous analysis, when the Co doping concentration $x > 0.04$, metallic Co clusters appear in the films, and the content gradually increases. However, the M_s shows the opposite trend, which helps to exclude the possibility that the observed ferromagnetism originates from the metallic Co clusters. The XPS results suggest that there exist a certain number of oxygen vacancies in the films, which strongly correlates with the formation of bound magnetic polaritons. Nonmonotonic behavior of M_s should be related to the existence of doped Co ions, which also affects the number of bound magnetic polaritons and thus alter the strength of ferromagnetic ordering. Fig. 7(b) displays the resistivity as a function of temperature for the studied films. The resistivity of films is measured at the temperature range from 10 to 300 K. Obviously, all the films exhibit clear conductive properties of semiconductors with negative temperature coefficients. Fig. 7(c) displays the variation of resistivity and carrier concentration with Co

doping at room temperature. The resistivity firstly increases to the maximum value at $x = 0.07$ and then gradually decreases with Co doping. It can consider that for the films with $x \leq 0.07$, Co elements exist mostly in the +2 valence state in the lattice. As the Co concentration increases, more and more Co^{2+} replaces the In^{3+} sites to compensate for the electrons provided by the oxygen vacancies, leading to a gradual increase of resistivity. And when the Co concentration further increases ($x > 0.07$), the resistivity gradually decreases, which is due to that the films with high x are gradually precipitated by metallic Co clusters, which makes the percentage of substitutional Co ion decrease, resulting in a weakening of compensation effect.

Electron transport behavior of semiconductor is important for guiding the development of new spintronic devices. Generally, there are three classical transport theories in semiconductors: (1) Mott variable-range hopping (MVH) model, $\rho \propto \exp(T_0/T)$, which is suitable for the system where the Coulomb interaction is neglected and carrier density of states (DOS) can be regarded as a constant [36,37]. (2) Efros variable-range hopping (EVH) model, $\rho \propto \exp(T_{\text{ES}}/T)^{1/2}$, which considers the Coulomb interaction between long-range charges, and is suitable for systems in which the Coulomb interaction between carriers is strong and carrier DOS decays exponentially [38]. (3) Hard gap hopping (HGH) model, $\rho \propto \exp(T_H/T)$, which is suitable for systems where the DOS has a constant band gap at the Fermi surface. Subsequently, the resistivity-temperature (ρ - T) curves of films are fitted individually using the models discussed above. Obviously, only the experimental curves are best fitted by the equation: $\ln \rho = \ln[A \exp(-BT) + C \exp(-DT^4)]$, where $A \exp(-BT)$ represents the Mott variable-range hopping and $C \exp(-DT^4)$ represents the Hard gap hopping, as shown in the solid line of Fig. 7(d). The A and C values are the constant, B is the temperature-dependent coefficient, $B^4 = T_0$ and the D value is the hard

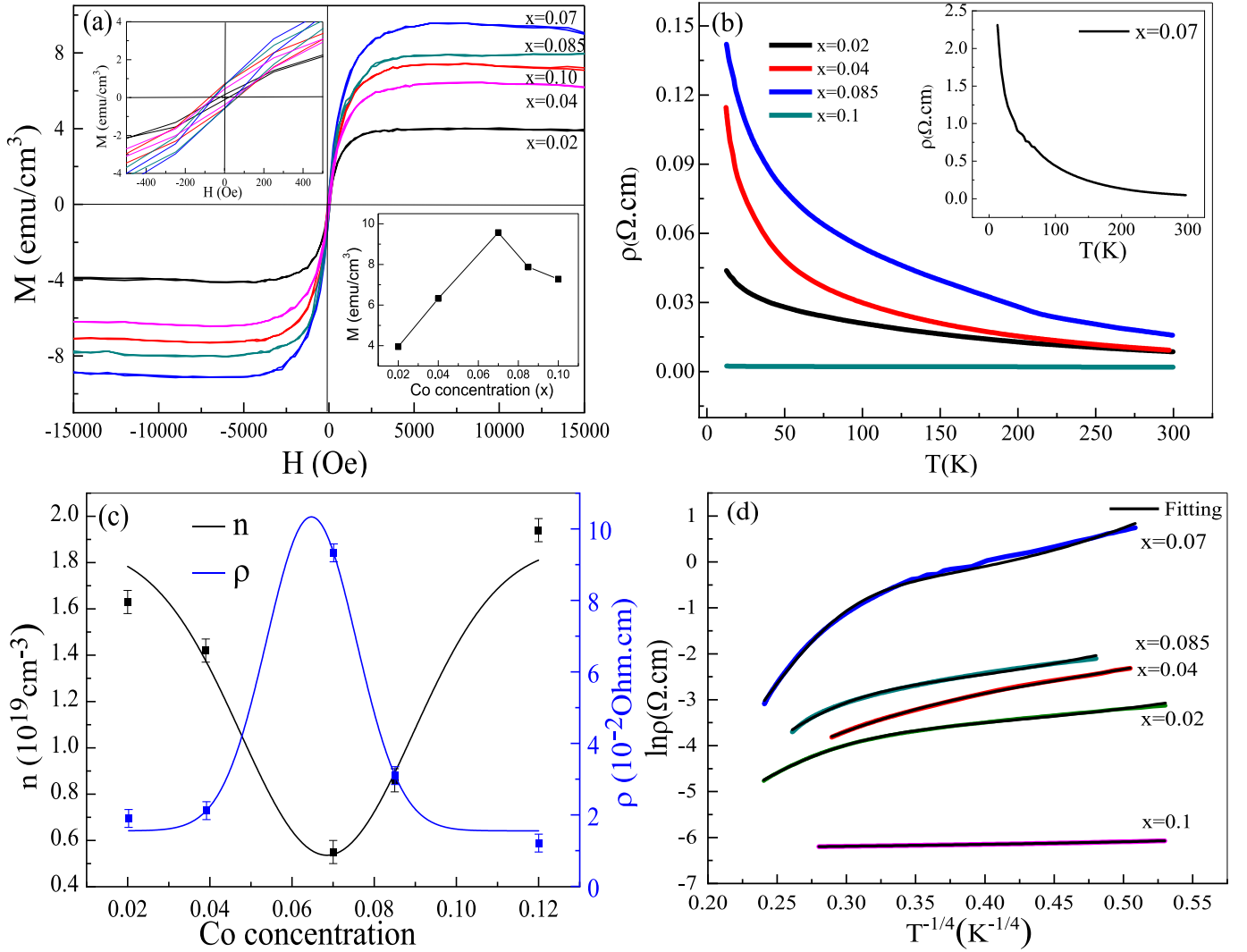


Fig. 7. (a) Hysteresis loops of $(\text{In}_{0.97-x}\text{Co}_x\text{Mg}_{0.03})_2\text{O}_3$ films at 300K, the top inset is the magnified loops and the bottom inset is the M_s as a function of Co concentration. (b) The resistivity as a function of temperature. (c) The resistivity and carrier concentration as a function of Co concentration x . (d) The relationship between $\ln\rho$ and $T^{-1/4}$.

band energy.

The parameter k_{Fl} , which is described by the formula: $k_{Fl} = \hbar(3\pi^2)^{2/3}/(e^2\rho n^{1/3})$, can well characterize the transport properties of carrier. The calculated values of k_{Fl} shown in Table 2 are less than 1 except for the concentration $x = 0.1$, and the $\ln\rho \cdot T^{-1/4}$ can well maintain the linear relationship at low-temperature range, strongly implying that the transport mechanisms of films are measured by MVH behavior with strong localization. Therefore, the carrier-mediated magnetic mechanism can be excluded. At the same time, it can be calculated the local radius of MVH: $\xi = (2\beta/N_d)^{1/3}(T_v/T_0)^{1/4}$. Where, the N_d , T_0 and T_v are

Table 2

The resistivity ρ , carrier concentration n , mean free path l , characteristic hopping temperature T_0 , localization radius ξ , hard-band energy (E_d) and onset temperature of hopping (T_v) for the $(\text{In}_{0.97-x}\text{Co}_x\text{Mg}_{0.03})_2\text{O}_3$ films.

Samples	$\rho(10^{-2} \Omega \text{ cm})$	$n(10^{19} \text{ cm}^{-3})$	k_{Fl}	l (nm)	T_0 (K)	ξ (nm)
$x = 0.02$	1.90	1.63	0.82	1.04	47646	1.76
$x = 0.04$	2.12	1.42	0.78	1.02	418498	1.23
$x = 0.07$	9.33	0.55	0.24	0.43	167962	1.17
$x = 0.085$	3.10	0.86	0.62	0.98	497016	1.28
$x = 0.1$	1.21	1.94	1.21	1.45	65536	1.58

the carrier concentration, characteristic hopping temperature associated with carrier hopping probability and the initiation temperature of MVH, respectively [39]. Clearly, the T_0 increases and then decreases with Co doping, as shown in Table 2. On the contrary, the local radius ξ of MVH in the films exhibits a opposite behavior. For the films with low concentration of doping ($x \leq 0.07$), the decrease of the local radius may enhance the local effect ξ of carriers and cause the contraction of the variable-range hopping region, which to a certain extent makes the crossover between bound magnetic polaritons less likely and improves the stability of BMP. When $x > 0.07$, the metal Co singlet scatters the electrons and reduces the electron migration in the “TM- V_O -TM” structure, further reducing the range of individual bound magnetic polaritons. Its effect on the ferromagnetism of the system will be further discussed in the following analysis.

Fig. 8 shows the field dependence of magnetoresistance (MR) with $x = 0.02, 0.04$ and 0.085 under various temperature. Here, the MR is defined as $MR = \rho_H/\rho_0$. As is well known, there exist two types of MR, namely, the positive magnetoresistance (PMR) and negative magnetoresistance (NMR). Generally, for the films in which the variable range hopping conductivity mechanism plays a dominant role, contraction of the carrier wave function occurs under the action of a strong magnetic field, triggering a positive magnetoresistance. Specifically, the contraction of the wave function indirectly leads to a decrease in the

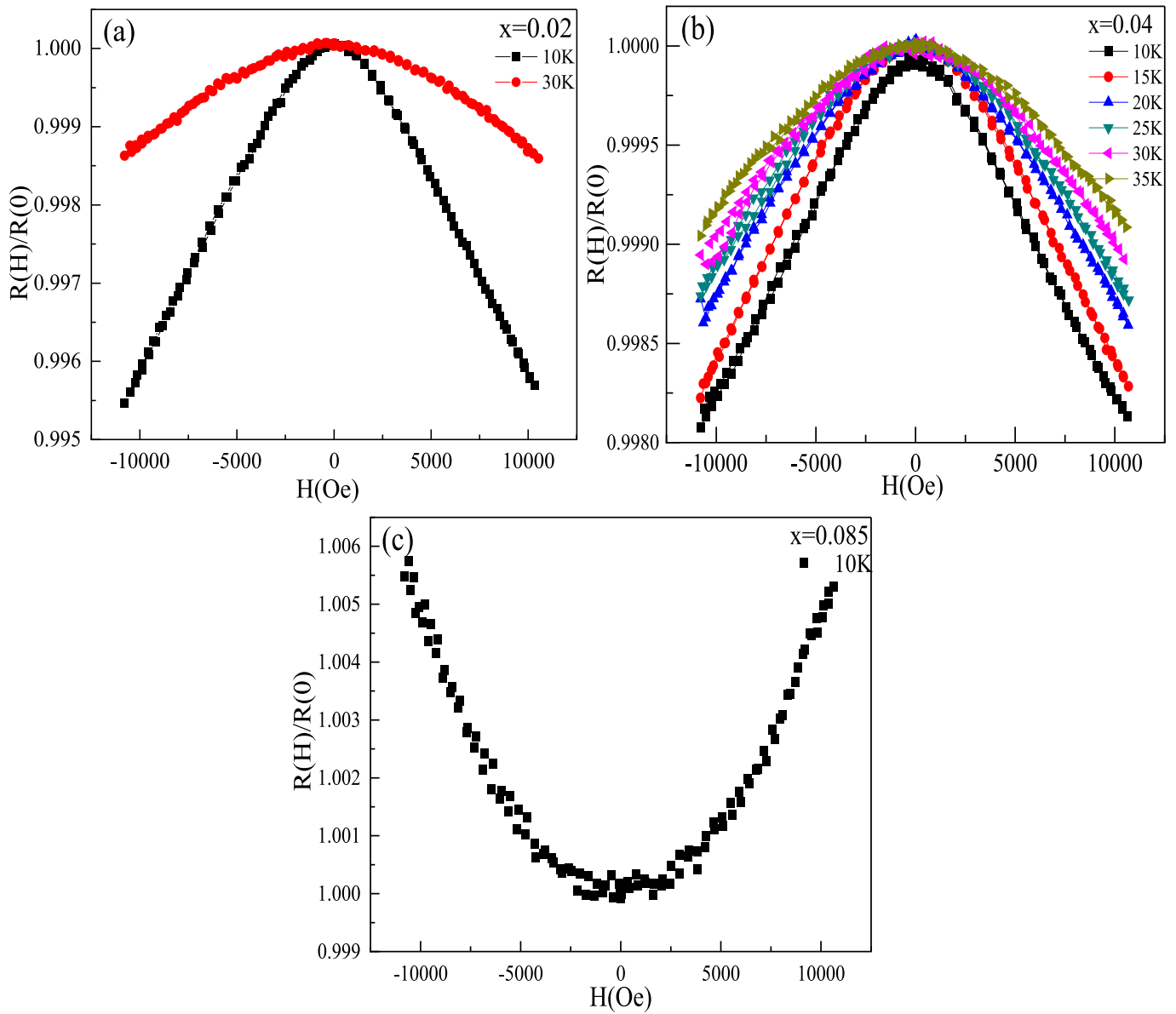


Fig. 8. Field dependence of magnetoresistance (MR) with (a) $x = 0.02$, (b) 0.04 and (c) 0.085 under various temperature.

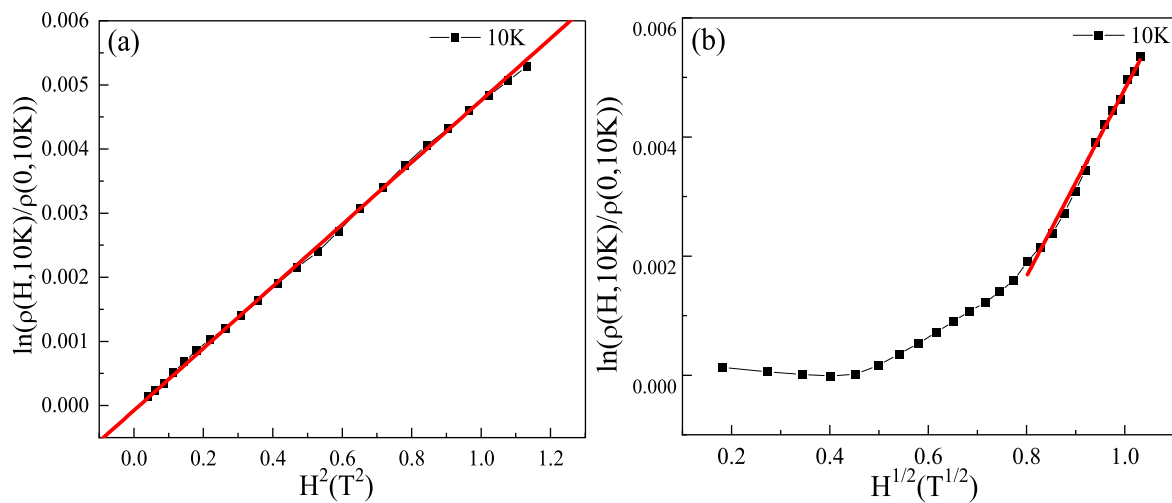


Fig. 9. (a) H^2 and (b) $H^{1/2}$ for the $(\text{In}_{0.885}\text{Co}_{0.085}\text{Mg}_{0.03})_2\text{O}_3$ film. Solid red lines are guides to the eyes.

chance of electrons jumping between the two states and causes an increase in the resistance. For the doped systems with s-d or p-d exchange, the carriers are surrounded by the polarized electron cloud of magnetic ions. When the systems are subjected to an applied magnetic field that aligns the spin direction of the magnetic ions, the migration of carrier becomes easier, which leads to a decrease in resistance, namely the NMR effect. As can be seen in Fig. 8, the films with $x = 0.02$ and 0.04 exhibit NMR behaviors at all temperatures. On the contrary, as the doping concentration increases to $x = 0.085$, the film exhibits the PMR behavior. Molnar et al. [40] proposed the bound magnetic polariton model to explain the NMR phenomenon of films. Since the fitting of ρ - T curves in Fig. 7(d) has suggested that combination of MVH and GHG models can well explain the transport behaviors. Therefore, it is preferred to use the wave function contraction model mentioned above to explain the PMR phenomenon. In the hard gap hopping region, the wave function contraction model can be expressed as:

$$\ln \frac{\rho(H)}{\rho(0)} \propto \frac{H^2}{T^3} \quad (1-1)$$

As shown in Fig. 9(a), the H^2 in the low field is fitted using Equations (1-1) and a good fit can be obtained. Also, for the study of MR behavior at high fields, a similar equation is proposed by Tian et al.:

$$\ln \frac{\rho(H)}{\rho(0)} \propto \frac{f(H)}{T^3} \quad (1-2)$$

where the $f(H) \sim H^{1/2}$ can well describe the experimental results, as plotted in Fig. 9(b). This strongly demonstrate that the PMR originates from the wave function contraction effect under the magnetic field.

Regarding the magnetism origin of doped $\text{In}_{0.97-x}\text{Co}_x\text{Mg}_{0.03}\text{O}_3$ films, double exchange interaction model, RKKY exchange interactions mediated by conduction electrons, mean-field Zener model, and BMPs mechanism have suggested. The structural and valence analysis of $(\text{In}_{0.97-x}\text{Co}_x\text{Mg}_{0.03})_2\text{O}_3$ films prove the substitutional form of Co ions in the +2 valence form with low Co concentration. While the high Co concentration doping ($x > 0.04$) results in the forming of Co metal clusters. As the Co concentration increases, the M_s firstly increases and then decreases with Co doping. However, the carrier concentration exhibits a opposite behavior. These prove that the carrier-mediated magnetic mechanism is not main reason of observed ferromagnetic ordering and the metallic Co clusters should be superparamagnetic behavior with very small size, which is consistent with the results reported by Vargas et al. [41]. Structural defects are an important factor for achieving ferromagnetic coupling in oxide-based magnetic semiconductors [42–44]. It is noted that there exist a certain number of oxygen vacancy defects in the films according to the results of XPS. These oxygen vacancy defects capture electrons to form the structure of $\text{In}^{3+}\text{-V}_\text{O}\text{-Co}^{2+}$. Then the electrons are transferred within such structures, and the metal ions spin isotopically aligned to trigger local magnetic ordering, forming bound magnetic polarons, which then overlap each other and exhibit ferromagnetic ordering on a macroscopic scale. In the analysis of the decomposition pattern of the Co 2p_{2/3} core level XAS spectra, the percentage of substitutional Co concentration gradually increases with Co doping, and the number of oxygen vacancies also gradually increases. This corresponds to the increase of number of bound magnetic polarons, which explains the gradual increase of M_s in the films with low Co concentration. When the Co concentration further increases, the content of metallic Co clusters also gradually increases, and the electrons captured by the oxygen vacancies are severely scattered and restricted in the $\text{In}^{3+}\text{-V}_\text{O}\text{-Co}^{2+}$, which reduces the range of bound magnetic polarons, and causes the decrease of M_s .

4. Conclusions

A series of $(\text{In}_{0.97-x}\text{Co}_x\text{Mg}_{0.03})_2\text{O}_3$ films with varying Co concentration are prepared by RF magnetron sputtering. The structure and

valence analysis show that the films exhibit good crystal orientation with clear/uniform stripe spacing, and the doped Co atoms replace for the In_1 sites of In_2O_3 lattices in the +2 valence form with $x = 0.02$. On the contrary, the films with higher Co doping concentration ($x > 0.02$) form a few metallic Co clusters. All the films exhibit n-type semiconductor conductive properties with strong localization of carrier, which are proved by the Mott VRH conductivity mechanism in the low temperature region. The films with Co concentration $x = 0.02$ and 0.04 only exhibit the negative magnetoresistance (MR) character. With increasing Co concentration to $x = 0.085$, the positive MR character is observed. All the films exhibit room temperature ferromagnetism and the M_s increases and then decreases with the increase of Co concentration. The room-temperature ferromagnetism is closely related to the bound magnetic polaritons induced by oxygen vacancies, and not the superparamagnetic metallic Co clusters.

CRediT authorship contribution statement

Jiashuo Yan: Writing – original draft, Investigation. **Cunquan Li:** Writing – original draft, Methodology, Investigation, Data curation. **Huijuan Sun:** Methodology, Investigation. **Yukai An:** Writing – review & editing, Resources, Investigation, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Acknowledgments

This work was supported by Tianjin Natural Science Foundation of China (Grant No. 17JCYBJC17300), Tianjin Graduate Research Innovation Project (Grant No. 2022SKY173), the Academic Research Projects of Beijing Union University (No. ZK70202007) and R&D Program of Beijing Municipal Education Commission (No.KM201911417002).

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